

HND Curriculum and module specifications in Electrical Engineering Technology

PROGRAMME: HND ELECTRICAL/ELECTRONICS TECHNOLOGY			
Course: TESTING METHODS AND RELIABILITY (Condition for Monitoring)			
Course Code:EEC 324		Contact Hours: 2/0/0	
Course Specification: Theoretical Content			
General Objective 1.0: Understand the basic terms and relationships involved in reliability studies			

Specific Learning Outcome:		Teachers Activities		Sources	
1	Explain the importance of reliability with respect to electrical/electronics	State practical applications and give examples	Text books	$R = e^{-\lambda t}$, $Q = 1 - R$, $R + Q = 1$, Where R = Reliability and Q = unreliability Sketch and label the bathtub diagram (i.e graph of failure rate against time) Explain the characteristic failure curves in the bathtub diagram. State the probable causes of failure in each of the regions of the bathtub diagram Sketch the wearout failure curve Interpret the wearout failure curve using normal (Gaussian) distribution. Determine the failure rate for a unit from the failure rates of its constituent parts using the relationship: Overall failure rate = basic failure rate λ x No. of similar parts weighing factor (environmental) x weighing factor (rating) x weighting factor (Temperature).	General Objective 2.0: Appreciate the concept of reliability prediction
	2 Define the terms: reliability, failure, item, mean-time to failure (MTTF) mean-time-between failures (MTBF) 3 Explain the meaning of the following types of failure misuse, inherent weakness, sudden, gradual, partial, catastrophic and degradation failures (wearout) 4 Differentiate between instantaneous and proportional failure rates. 5 Explain the reliability equations and the related curves when λ is constant, i.e. 6 $R = e^{-\lambda t}$, $Q = 1 - R$, $R + Q = 1$, 7 Where R = Reliability and Q = unreliability 8 Sketch and label the bathtub diagram (i.e graph of failure rate against time) 9 Explain the characteristic failure curves in the bathtub diagram. 10 State the probable causes of failure in each of the regions of the bathtub diagram 11 Sketch the wearout failure curve 12 Interpret the wearout failure curve using normal (Gaussian) distribution. 13 Determine the failure rate for a unit from the failure rates of its constituent parts using the relationship: Overall failure rate = basic failure rate λ x No. of similar parts weighing factor (environmental) x weighing factor (rating) x weighting factor (Temperature).				

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 - find 15-12 LT2

	Explain the basic probability rules in relation to Reliability calculation; 1. Multiplication and addition rules The Binomial probability distribution^{2,2} Determine mathematical expression for the reliability and MTBF of series systems. Derive mathematical expression for the reliability and MTBF of systems. Determine the reliability and MTBF of series, parallel and non-series, non-parallel systems. Explain the meaning and significance of redundancy Differentiate between active and passive (standby) redundancy. Solve simple problems relating to active and passive redundancy. State practical application of active and standby redundancy		
6	General Objective 3.0: Understand the causes and remedies of component failure Explain the cause of failure due to environmental factors i.e effect of temperature, humidity, atmosphere pressure, chemical content and radiation. Explain the cause of component failure due to operating stresses i.e effect to operating voltage, current and frequency. Explain other causes of component failure due to mechanical stresses such as shock vibration and friction. State specific methods of dealing with environmental problems Explain 'Deteriorating' as a method of dealing with failure problems caused by operational stresses, i.e. apply the Arrhenius law (the fifth power law) to illustrate derating.	Explain by using practical application	Components of electronics e.g Resistors
	General Objective 4.0: Understand the basic principles of maintainability		

4.1	Define the term "maintainability";	State practical applications of maintainability	Text books
4.2	Explain the importance of maintainability in relation to reliability.		

7-9	4.3 Define the following terms: a. Utilization factor b. Availability c. Unavailability d. Repairability 4.4 Explain the concept of preventive and corrective maintenance 4.5 State the factors affecting maintainability 4.6 Explain the methods of improving maintainability 4.7 Illustrate graphically the relationship between cost and equipment reliability 4.8 Explain the concept of failure reporting		
General Objective 5.0: Appreciate the purpose of specifications			
	5.1 Define the term "specifications" 5.2 State the aims and uses of specifications 5.3 List typical items of information that should be included in specifications 5.4 Illustrate 5.3 with examples of specifications for typical measuring equipment	Use question and answer to explain	Text books
General Objective 6.0: Appreciate the need for testing; types of test carried out and the purpose of testing			

12-15	6.1 Explain the meaning of the following terms: a. reliability demonstration test b. reliability acceptance test c. calibration test d. non-destructive test e. testing for packaging and transport f. identification test g. preproduction test Give an example of each test stated in 6.1 above Explain the relationship between prototype items of equipment. State the reason for producing prototype items of equipment. Explain the necessity for pre-production testing Explain the different approaches needed when testing prototype, small batch quantities and large batch quantities	State typical examples of the various test	
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ASSESSMENT: Course work 20%; Course tests 20%; Examination 60%.



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Q.1. MAINTENANCE

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MAINTAINABILITY (2000)

Definition

Maintenance is a combination of any action carried out to repair an item in or restore it to normal operational standard.

Adv: Maintainability is the ease and speed with which a failed item can be repaired and brought back into service.

Maintainability is the probability that a unit or system will be restored to operational efficiency within a given period of time when the maintenance prescribed in accordance with its pertinent to maintain that the achievement of high maintainability is dependent on some factors: such as type of system, location, efficiency and skill of the maintenance crew and provision of spare parts etc.

What is the effect of principal objectives of maintenance?

(4.2) Some important terms related to maintainability

(1) Maintenance action: The prescribed operation to repair a failed item.

(2) Mean time to repair (MTTR): Is the mean of the time required to perform maintenance actions. A typical sequence



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of maintenance tasks comprises fault detection, fault isolation, fault correction, repairing, removal, replacement assembly, and alignment or adjustment, and check out.

$$\text{Mathematically MTR} = \sum n_i r_i t_{m_i}$$

$$\sum n_i r_i$$

where, n_i = quantity of similar parts
 r_i = parts failure rate.

t_{m_i} = predicted maintenance operation time.

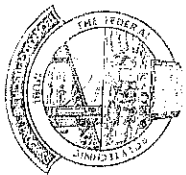
~~(ii) Maintenance time constraint:~~

~~(iii) Maintenance action (repair) time rate: (iv)~~
The number of maintenance action that can be carried out on a particular item per hour or average number of item (equipments) which can be restored if normal working condition per hour, assuming that there is no working hrs repair jobs.
~~Maintenance action~~

$$\text{Mathematically, } M = \frac{1}{MTR} \quad \text{--- (4.2)}$$

(iv) Maintenance time constraint: t_{m_i} - It is the permissible repair time.

(v) Utilization factor (U): Utilization factor of a unit or system is defined as the ratio of the operating time (t_{op}) to the sum of the maintenance time (t_m), idle time (t_{id}) which may occur also completion of the maintenance and operating time.
The utilization factor can be expressed as:



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$U = \frac{\text{top}}{\text{top} + \text{fid} + \text{top}}$

(4.8)

(11) Availability (A) is the probability that an item will perform its required function at a stated instant of time or over a stated period of time.

Mathematically, $A = \frac{\text{top}}{\text{top} + \text{fid} + \text{top}}$

(4.9)

(Application factor is measurement, Unm)

On the other hand, read our knowledge of MTBF & MTTR

same top = MTBF

bottom = MTTR

$A = \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}}$

(4.10)

(12) Unreliability (R) is the complement of Availability

$R = 1 - A$

$\approx 1 - \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}}$

$\frac{\text{MTTR}}{\text{MTBF} + \text{MTTR}}$

$\approx \frac{\text{MTTR}}{\text{MTBF} + \text{MTTR}}$

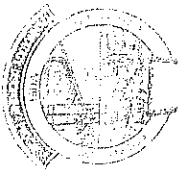
$\frac{\text{MTTR}}{\text{MTBF} + \text{MTTR}}$

$R = \frac{\text{MTTR}}{\text{MTBF} + \text{MTTR}}$

(4.11)

Example (1)

An electrical generating set designed for continuous operation fails three in a period of 123 days. The stated



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time for repairs during the past 3d of 15
3 days - Determine the parameters:

(i) MTBR (days), (ii) MTTR (days) (iii) Availability
to Unavailability.

Soln

(i) Total operating time = $(123 - 3) \text{ days} = 120 \text{ days}$
Total number of failures = 2

$$\therefore \text{MTBR} = \frac{120}{2} = 60 \text{ days}$$

(ii) MTTR = Total time for repairs
Total number of failures
= 36 day or $1\frac{1}{2}$ day

(iii) Availability = $\frac{\text{MTBR}}{\text{MTBR} + \text{MTTR}}$

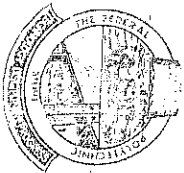
$$= \frac{60}{60 + \frac{36}{2}} \times 100\%$$
$$= 97.6\%$$

(iv) Unavailability = $1 - A$

$$= 1 - 0.976$$
$$= 0.024$$

Example (2)

4.6 CLASSIFICATIONS OF MAINTENANCE
It is relevant to discuss briefly
each and all forms of maintenance
plan below.



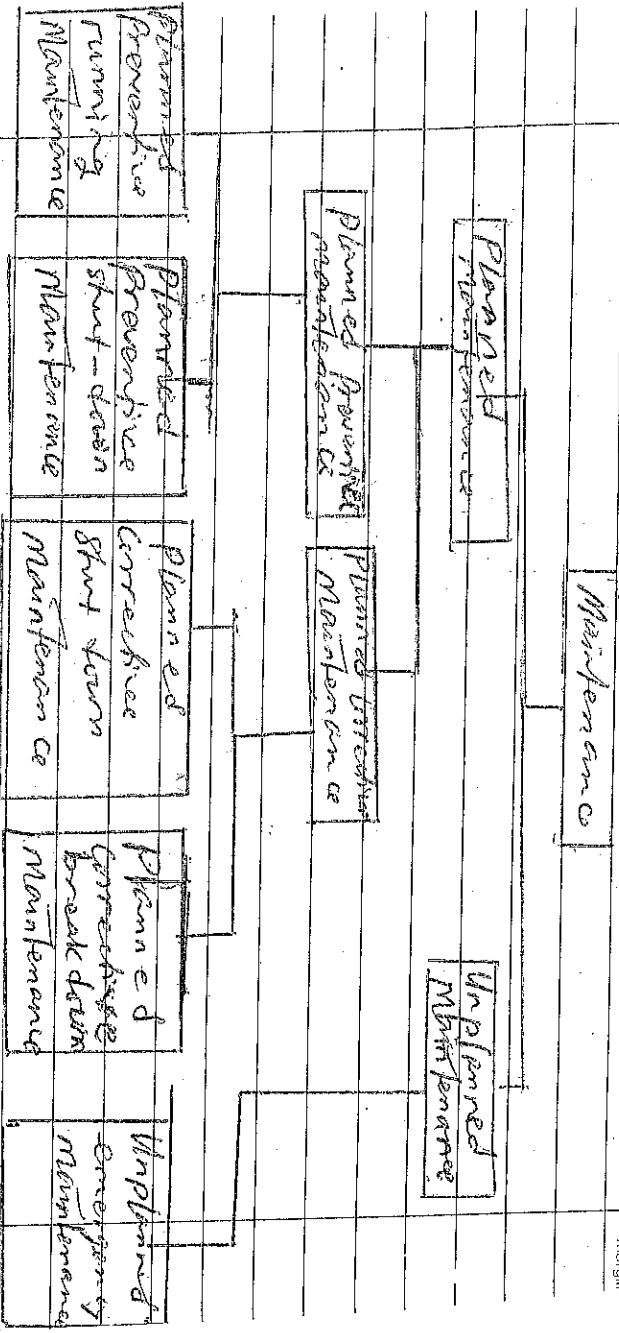
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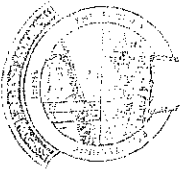
(i) Planned Maintenance: Is the type of maintenance, where organised and carried out with forethought, control and record. It is not a specific type of maintenance, but represents the application of maintenance packed in a scientific manner.

(ii) Planned corrective Maintenance: It is normally engaged when an equipment has already failed and it is to be repaired and brought back to service. The two types of ~~planned~~ correct maintenance are:

(a) Shut-down Maintenance: Is often initiated by a sudden.

(b) Shut-down Maintenance: Is normally carried out when the equipment is out of service.

(c) Planned corrective break down m/c



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to often initiated by a sudden unexpected failure of an equipment.

(iii) Unplanned m/c: It is the type of m/c where not initially organised and not carried out while any trouble or control and records.

(iv) Pre-emptive Planned m/c: It is a maintenance in which components which are removed and replaced by new ones at a specific interval of time, in order to improve the efficiency.

4.7 Methods of Improving Maintainability
Some added factors of improving maintainability are:

(i) Provision of maintenance check point should easily accessible and hazard-free.

(ii) Provision of suitable tools and test equipment which are easily to operate.

(iii) Adequate training for maintenance personnel

(iv) Provision of well written reports and manual which can be easily ~~for~~ followed and understood.

(v) Provision of Adequate Spare parts

(vi) Provision of a Conducive environment for repairs and maintenance operations.

4.8 Relationship Between Reliability and Cost.



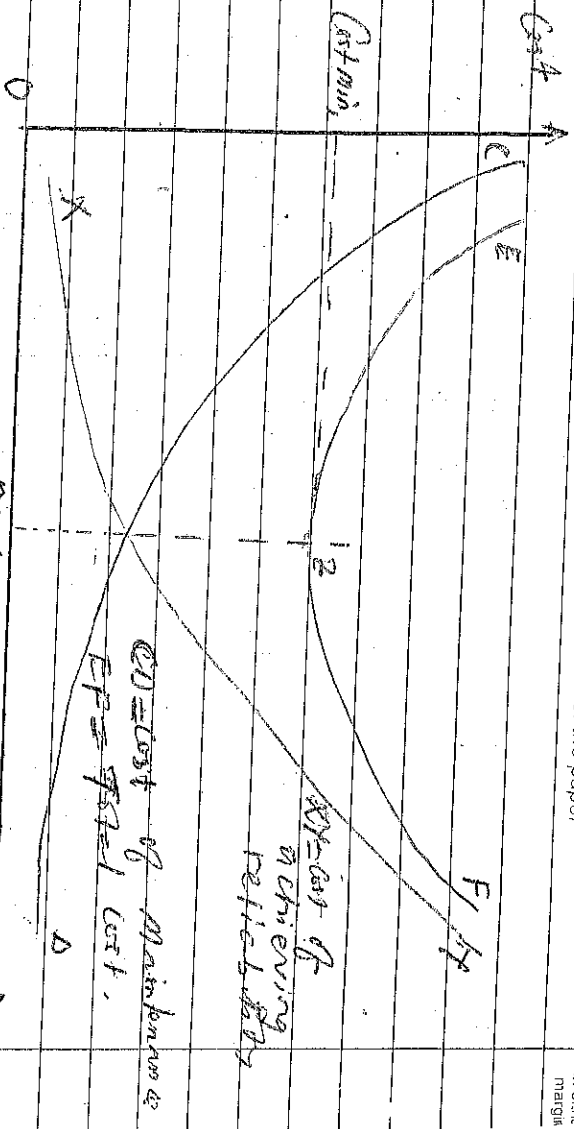
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Experiment	Reliability	V.O
Fig 4.5: Error/used relationship b/w reliability and Cost.		



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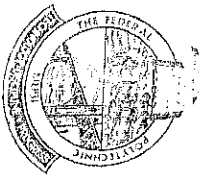
Secondly, ~~let~~ considering curve CD, when reliability is low, maintenance cost due to breakdowns are correspondingly high (region C). As the reliability improves, the cost of maintenance consequently decreases and approaches zero. (region D). As reliability approaches 1.00.

lastly, considering curve EF ~~and~~ $EF = XY + CD$. The point E - which gives an optimal value of reliability 'Ropt' and for which the overall cost per item produced is minimum.

~~After the above considerations~~ - Note that the above consideration is done only for academic purpose the case in actual situation which is usually complicated (difficult to determine the many values of achieving reliability and maintenance). Further, in an actual situation is always complicated that because of the following reasons:

(i) Company to find an assumption, it may not necessarily follow that reliability could be achieved only at the expense of more cost. This is so because improvement in design, use of better materials, improvement in quality control during production etc, may achieve improved reliability at little or no extra cost.

(ii) Maintenance cost are also not easy to determine. It is difficult to pre-determine



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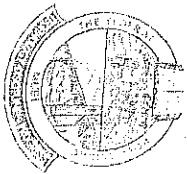
The repair cost of an equipment price one cannot predict all possible causes of equipment failure and resources that will be needed to effect repairs. (iii) Cost estimates are invariably mixed up with many factors, the practical implications of which may not be easy to determine.

4.9 Reporting Failures
Important of Failure Reporting.

Failure reporting is important because it is the means by which the designer becomes aware of the short coming in his design, so that he can adopt a corrective measure. The report failure report serves as a feedback to the manufacturer ~~to~~ in order to improve ~~the~~ reliability.

The following points are worthy of note in reporting failures:

- (i) Note only defects that actually lead to a breakdown should be reported. Usually rapid deterioration in a part, which may be revealed by normal maintenance procedures, is clearly a failure of the part even though replacement may be possible while the equipment is still functioning normally.
- (ii) Accuracy and speed are



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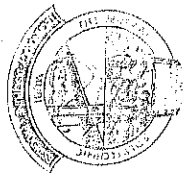
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essential features in reporting should the designer receive wrong information about the nature of the failure, reporting it could do more harm than good. Therefore, it follows that the nature of the failure must be accurately ~~described~~ diagnosed, the part affected has to be identified and the information incorporated in the report in a way most described/depict the true nature of the failure. These depends mainly on the technical & descriptive ability of the writer and of course the time he has available to complete the report.

(iii) Reporting failures should be seen and regarded as a two-way business. On the one hand, the user must appreciate that accurate reports are the most important method of improving reliability of an equipment after production. On the other hand, the designer/manufacture must realise that failure report provide the main source of information on which the reliability and improvement of future designs can be based. Therefore, the user and the designer must work hand in hand to achieve ~~the~~ maximum reliability.

(iv) Reporting failure is no formality



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Therefore, all those given the task
of reporting failures should realise
that reports should be completed
quickly and accurately.



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Ques Village: 20/250 or 100/1251 kms
50 or 60th
Consumption: 64%

(6) TESTING METHODS
INTRODUCTION

It is no simple matter trying to classify feeding. This is because the criteria for the classification are not so distinct and therefore not free from causing possible overlapping.

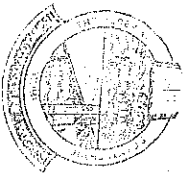
Test is something done to find out whether a thing is good, strong, efficient etc.

3 categories of feeding.

The three categories of feeding are called:

- Prototype feeding.
- Pre-production (Qualification) feeding
- Production feeding.

Apart from these, there exists reliability demonstration and acceptance testing which of course, may not be a mandatory process after the production of an item. Finally, there is also as is known as accelerated life testing. This actually does not stand on its own as a category of feeding but associated with any of the three stated above.



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6.2 Prototype Testing.

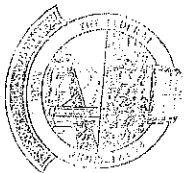
Before a design goes into production a prototype or few prototypes are first produced and are thoroughly tested to prove that a design will meet the specific condition. If the tests reveal any inadequacies in either performance or any other parameters, the design must be modified and the test repeated. This is desirable that prototypes are tested under the correct environmental conditions by competent staff who are independent of the designers. These test are made responsibility of competent staff in the reliability or the quality control department. As a result of prototype testing three types of errors may be revealed. They are:

(i) Design errors - These the prototype might have been produced according to drawing, but still it does not work satisfactorily.

(ii) Manufacturing errors - When the prototype has not been made correctly, in line with the design.

(iii) Parts fault - When individual component, parts of the prototype are found to be defective.

6.3 Pre-production (Qualification) Testing.
This type of test is aimed at ensuring that a production model fulfills all the



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Technical requirements stated in the Engineering specifications. Pre-production testing is normally done by built upon in order upon to ensure that the specification can be met under normal manufacturing conditions.

Pre-production testing comprises of the following:

- Performance test (at defined profit margins)
- Service test
- Reliability test
- Maintainability test
- Packaging and transport test (incline vibration and shock test)
- Physical test (to consider the positioning of operating levers and switches etc.

Below are some detail description of few of the test listed above:

* Service test: This is the test normally carried out on the production model under simulated environment conditions.

- Reliability test: The type of reliability test which is normally carried out on pre-production model concerns mainly reliability estimate. The test is done to determine the failure rate (MRR) or - Mean time between failure (MTBF) test will be necessary in checking the relative speed with it takes to

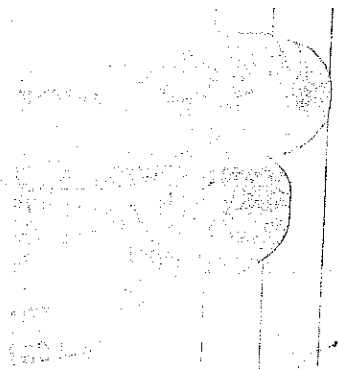
Non-destructive Testing

Rotation with sound, magnetic field, or liquid penetrant type of probing agents are used to non-destructive testing. Observation of the relative response of the probing agent helps in detection of a plan where size is measured

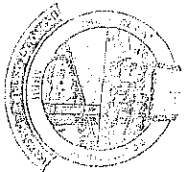
Destructive Testing

Tested in it permanent failure are done by a destructive test. Mechanical stress are

put on the components of
component supervise the
shares, when the company
is very reliable, because
feeding is very effective
for stable situation



Wang's Review
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same repair to be carried out on two alternative designs.

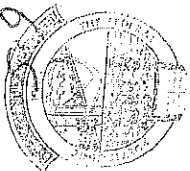
- Packaging & Transport Test.

In the sense in spending a lot in design and manufacture of a product without adequate regards for packaging and transport at identifying potential causes of defects during packaging, storage and transport - Ergonomics Test. This test is aimed at determining the effects of interface of a product with operators and maintenance personnel. Such test may reveal possible frequency of operation mistakes or the time required by the operator to carry out an operation on an equipment. The position and number of operating buttons and knobs constitute an important factor in determining an operator's convenience and accuracy in carrying out his job.

6. Production Testing.

The conformity of a finished product with equipment with specification - This test may involve some repeatability proving and determination of possible causes of early or late turn-in - failure. At this stage, failure may be attributed to component defect, production method etc.

7. Reliability Demonstration & Acceptance Testing.



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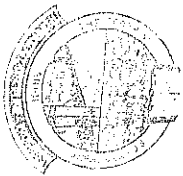
Reliability Acceptance Test

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This type of test is associated with the post-production stage. It should be intended that this is not a mandatory exercise, and when it happens it will be an option for the manufacturer and the customer.

Defect Reliability Demonstration Test is a test which is normally carried out as a manufacturer, having produced a certain equipment, wishes to demonstrate to a customer that its reliability is at least as good as he claims it to be. Conversely, a customer may receive a finished product untested and he decides to test it himself to verify whether or not the product satisfies the reliability figures as stated in the production specifications. In this case the customer will be said to be doing what is called a reliability acceptance test.

Accelerated life testing, accelerated life testing is done on some or a number of prototype products. These tests are usually employed in order to reduce the very long test times which are often necessary. The underlying idea is that the normal life cycle of the item concerned should be compressed into a very much shorter time for the purpose of the test. The



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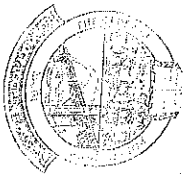
component or item is subjected to operate beyond normal limit. specified for voltage temperature, and clock rate the test for operating margins and the expose after failure.

6.7 Relationship of Test & Inspection
Various types of test can be carried out at different production stages have been thoroughly discussed. However, there is a tendency to confuse inspection with testing. Unless an effort is made to differentiate them.

Testing normally takes place at the end of each production stage, after prototype production, after pre-production stage and after final production stage. However, the inspection process may take place during and after each production stage. Inspection is usually directed at detecting defective items, further more, inspection is done mainly for

① If may provide information that may necessitate the adjustment of the production process in order to reduce the amount of production.

② To sort out the bad production and return items for pre-processing.
6-8 Relationship of Quality & Reliability
Quality can be defined as conformity to specifications or fitness for purpose.



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while reliability has to be
on design. It may be argued that
quality is a static feature while
reliability is a dynamic one.

e.g. Sampling Plans (Appropriate)
for both small and large batch
quantities.

We have discussed earlier that
it is unreasonable to carry out 100%
test on undisturbedly produced items (especially when the quality involved is
large) unless absolutely defect-free item
are needed and the are ready
to pay the cost in practice (instead
of embarking on inspection testing of
each item). Since not of acceptance
sampling is adopted. The types
of sampling methods commonly
embarked on are:

- (1) Single Sampling Plans
- (2) Double Sampling Plans

The above sampling methods
will be discussed vividly in the
classroom / lecture.

~~24/10/2020~~ } SVD
Date